The Geometry of Political Abstraction in Large Language Models

From diverse contexts to shared representations

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Motivation

LLMs encode linearly high-level concepts with heterogeneous domain-specific manifestations [e.g., Tegmark & Tegmark, 2024; Kim et al., 2020,].

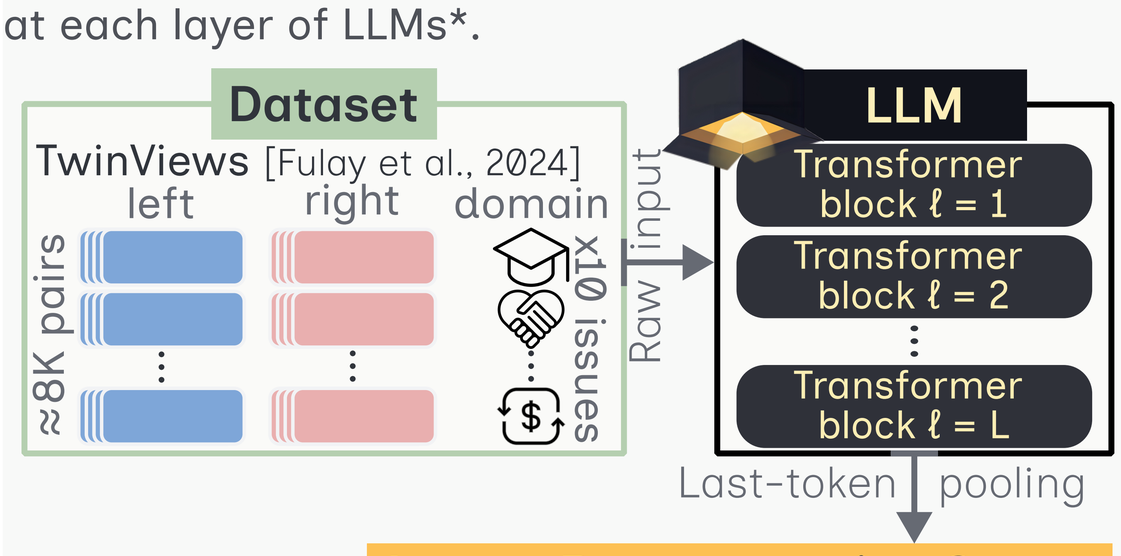
Abstraction

Def.: *A representational geometry supporting the generalisation of a variable across diverse conditions* [Bernardi et al., 2020; Courellis et al., 2024].

Yet, it remains unclear **whether their internal representations are organized in a form that supports *abstraction*.**

Experimental Framework

Investigation of **ideological abstraction across diverse political issues,** through latent representations extracted at each layer of LLMs*.

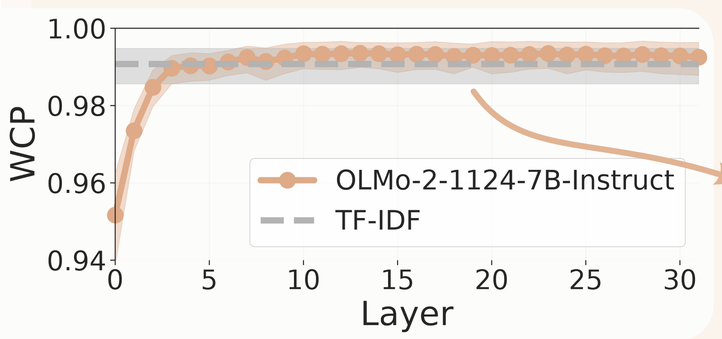


Probing Decoding Directions

Binary political leaning ("left"/"right") decoding through **domain-constrained SVCs** probes.

Within-Domain Performance (WCP)

RQ1: Can domain-specific ideological polarisation be linearly decoded?



$$\frac{1}{D}\sum_{i=1}^{D}\theta_{i\to i}$$

Yes, perfect in-domain linear separability!

... but does it reflect a collection of independent domain-specific features, or a unified overarching ideological abstraction?

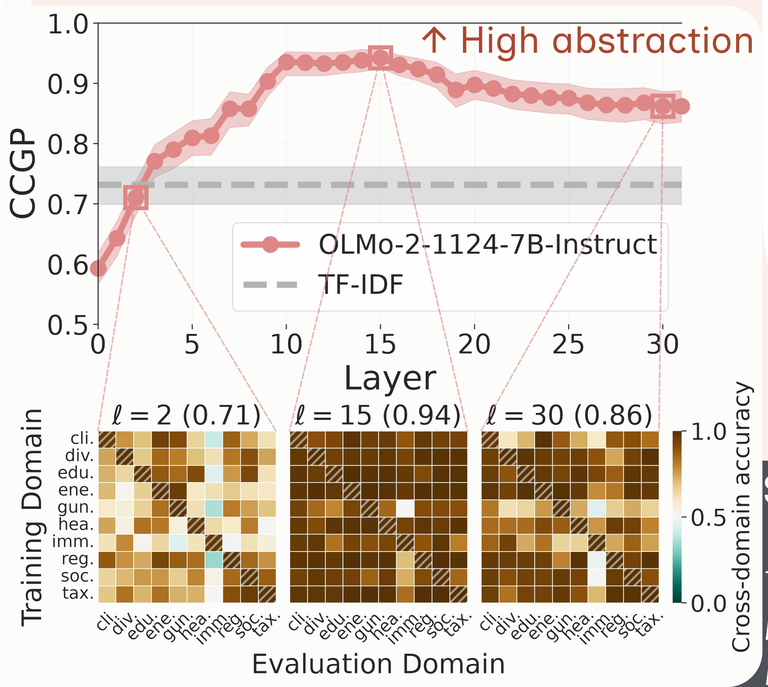
Cross-Domain Performance (CCGP)

RQ2: Do LLMs abstract ideological dimensions?

Cross-Condition Generalisation Performance (CCGP) tests semantic abstraction within representation spaces.

It measures the performance of probes trained on one domain when evaluated on other domains:

$$\frac{1}{D(D-1)}\sum_{1\le i\neq j\le D}\theta_{i\to j}$$



Effective Dimensionality

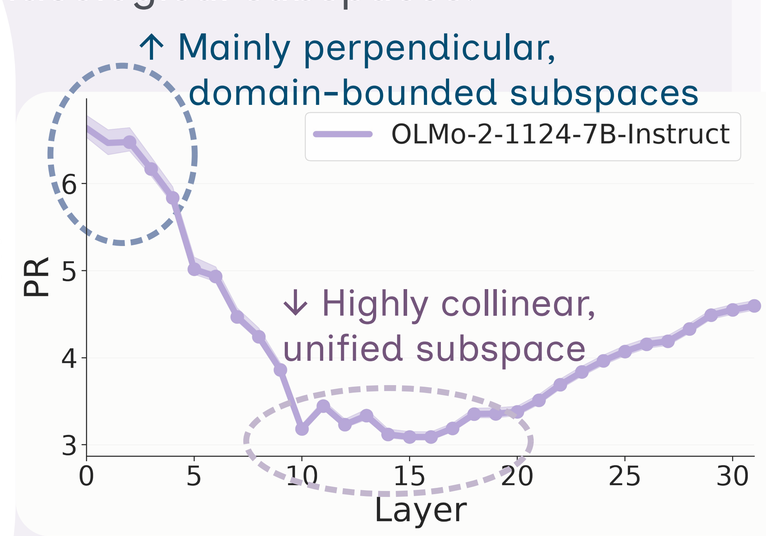
Analyse geometrical structure to **test if abstraction is supported by a compression** of political concepts, rather than decentralized, high-dimensional layout.

Participation Ratio (PR)

RQ3: How do LLMs geometrically organize ideological abstractions?

PR computed on domain-constrained ideological contrast vectors (stance-wise centroids differences).

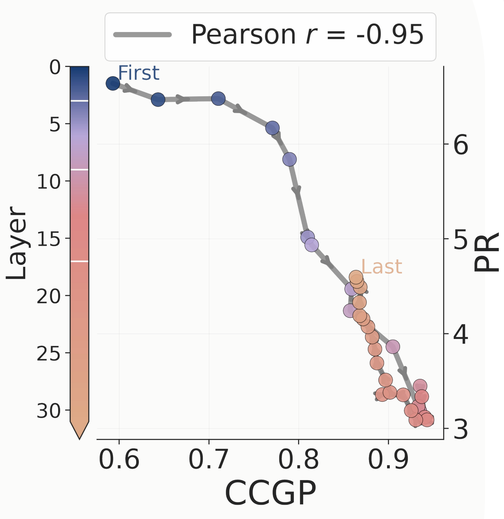
Provides a continuous estimation of effective dimensionality of the ideological subspaces.



Computational Arc

1. Lexical Buildup: Domain features processed independently.

2. Geometric Compression & Abstraction: Heterogeneous domains collapsed onto a low-dimensional subspace, with aligned contrast directions.



3. Task Decompression: Low-dimensional subspace gradually unpacked to feature-richer space.

d-noe/llm_political_abstraction

*Additional results in the article.

Selected References

[Marks & Tegmark, 2024] *The Geometry of Truth: Emergent Linear Structure in Large Language Model Representations of True/False Datasets.*

[Kim et al., 2025] *Linear Representations of Political Perspective Emerge in Large Language Models.*

[Bernardi et al., 2020] *The geometry of abstraction in the hippocampus and prefrontal cortex.*

[Courelis et al., 2024] *Abstract representations emerge in human hippocampal neurons during inference.*

[Fulay et al., 2024] *On the relationship between truth and political bias in language models.*